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(54) **NUCLEAR MAGNETIC RESONANCE APPARATUS**

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(57) **ABSTRACT**

A nuclear magnetic resonance apparatus including a magnetic field generator generating a magnetic field, a sample container held in the generated magnetic field and containing a sample to be analyzed, a probe that emits an electromagnetic wave at the sample and detects a nuclear magnetic resonance signal from the sample, a first protective cover surrounding a space where a strength of the generated magnetic field is equal to or greater than a first threshold, a second protective cover mounted outside the first protective cover and surrounding a space where the strength of the generated magnetic field is equal to or greater than a second threshold is weaker than the first threshold, an opening provided in the second protective cover, and a probe adjuster disposed outside the first protective cover and inside the second protective cover in vicinity of the opening to adjust a characteristic of the probe by an operator.

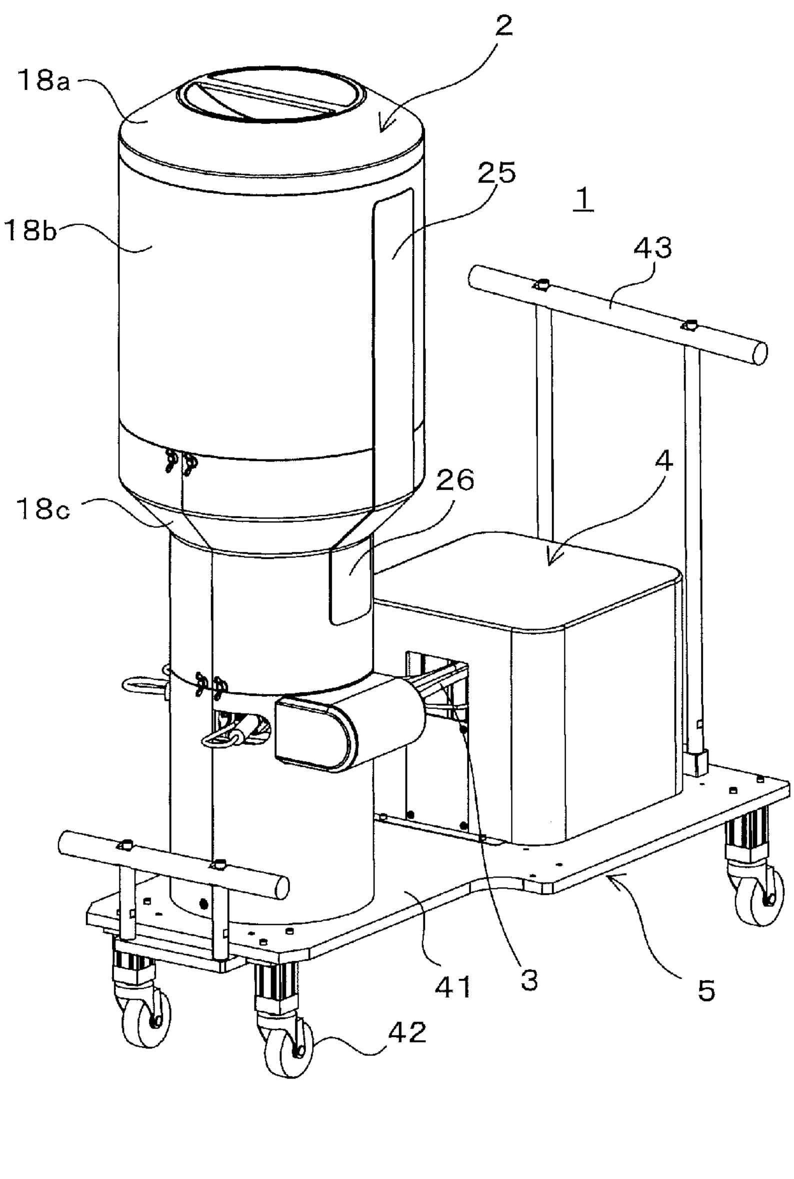


Fig. 1

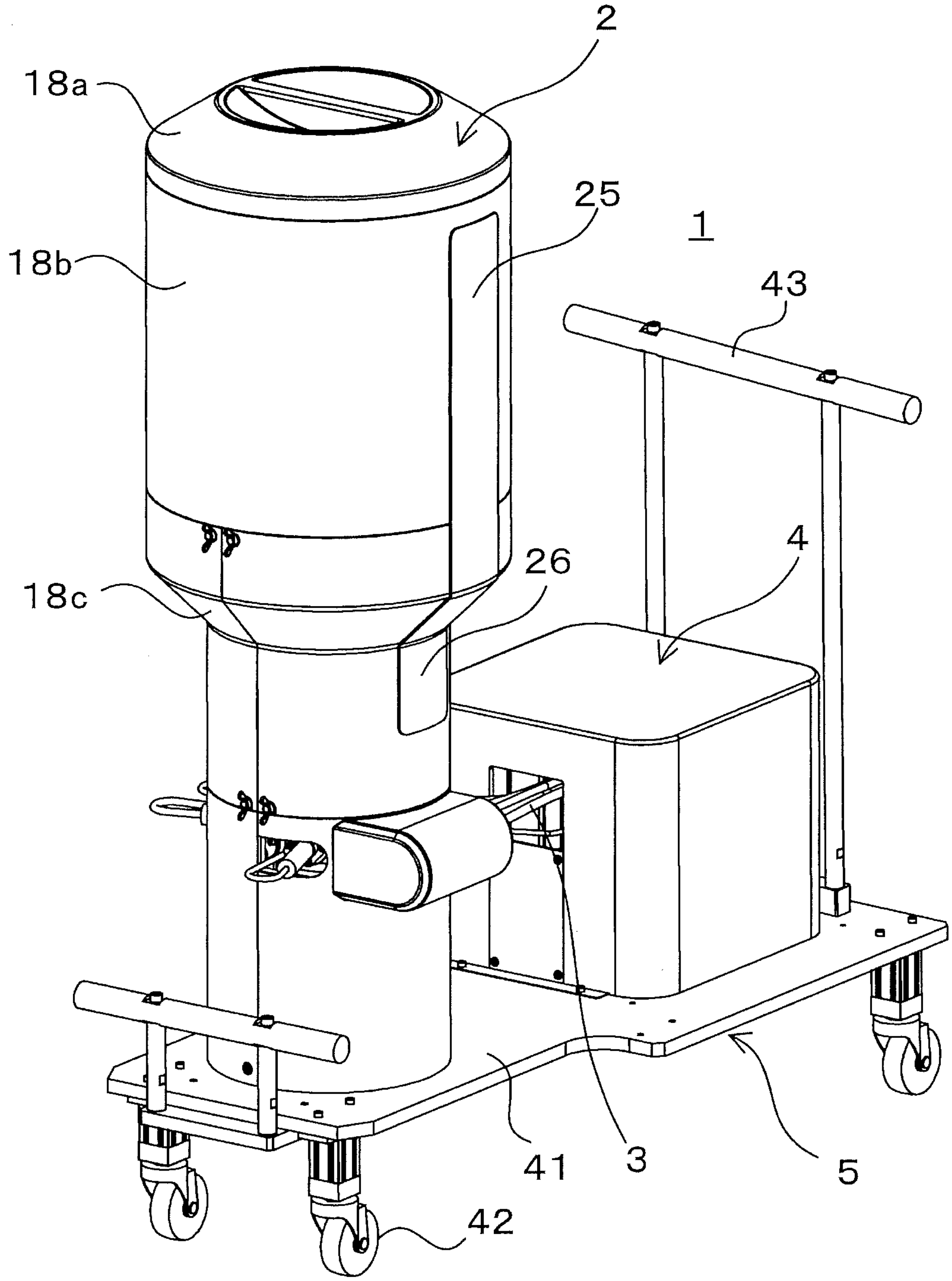


Fig. 2

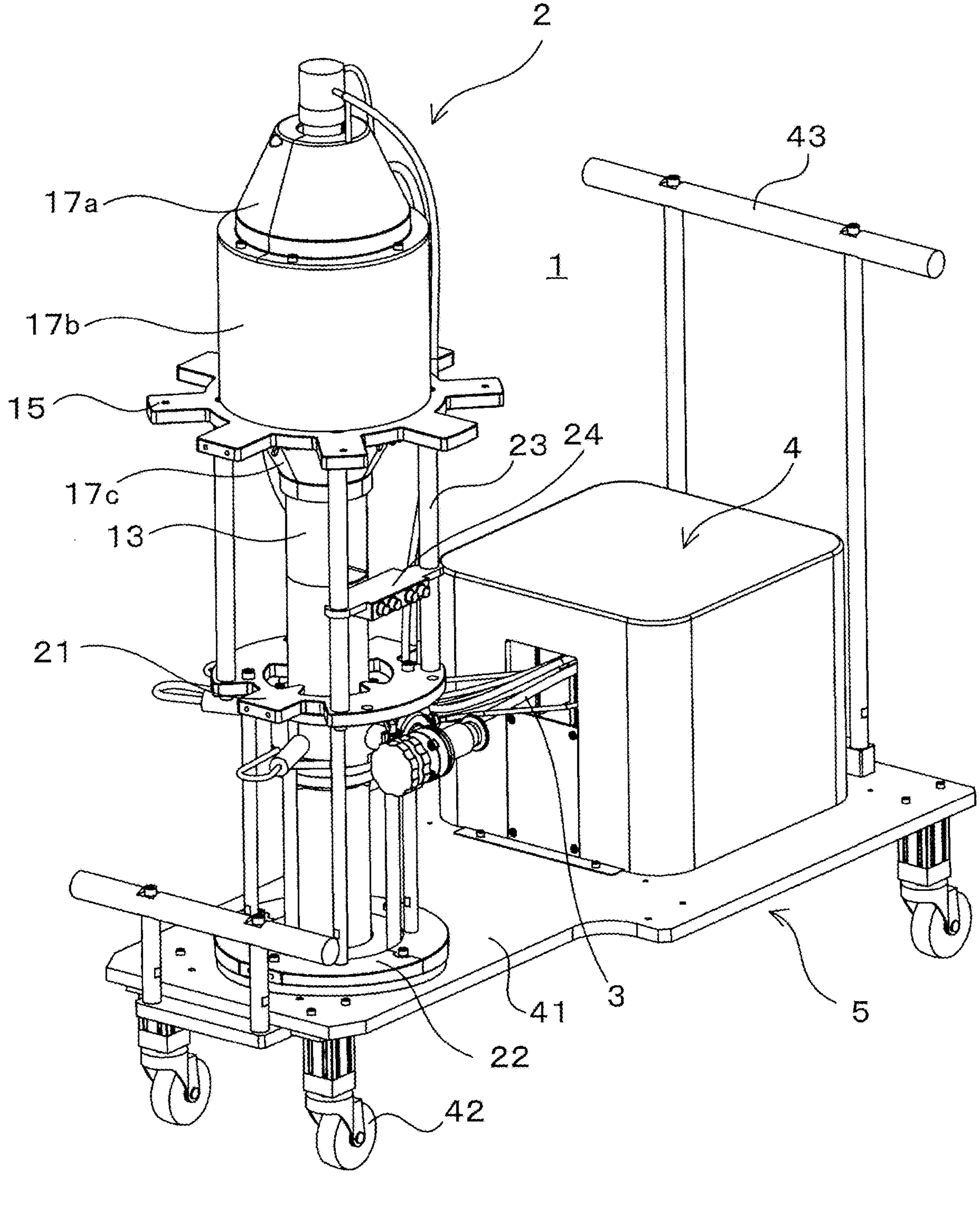


Fig. 3

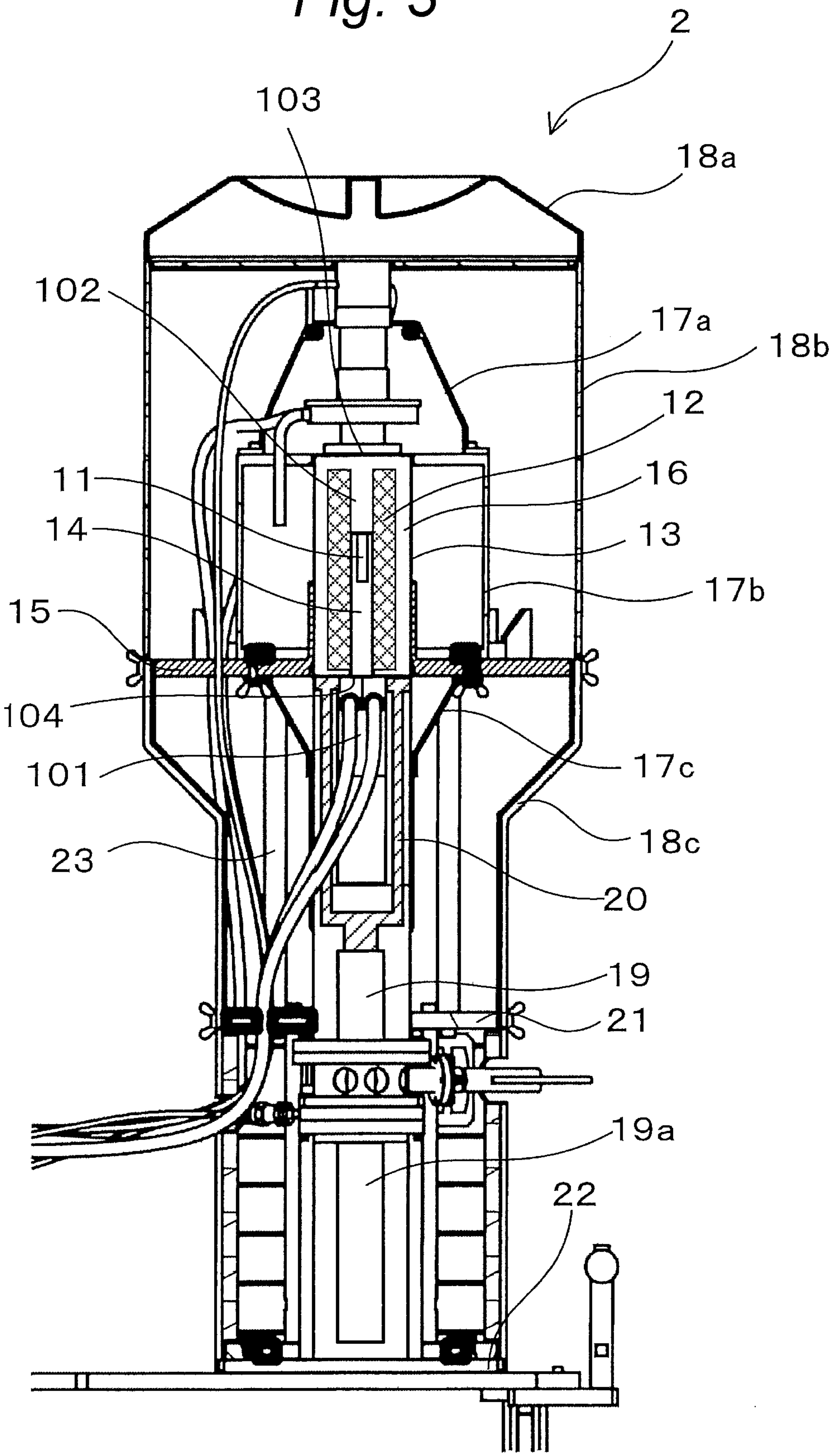

2
103
18a
102
17a
18b
12
16
11
13
14
17b
15
104
17c
101
18c
23
20
19
21
19a
22

Fig. 4

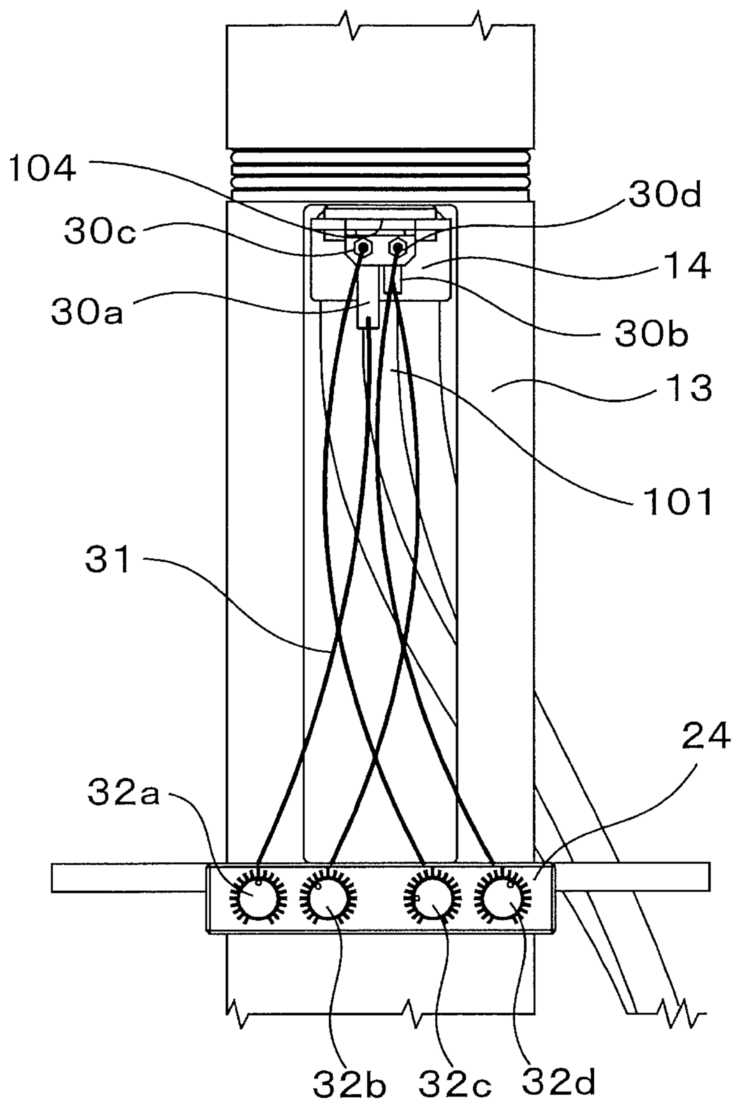


Fig. 5

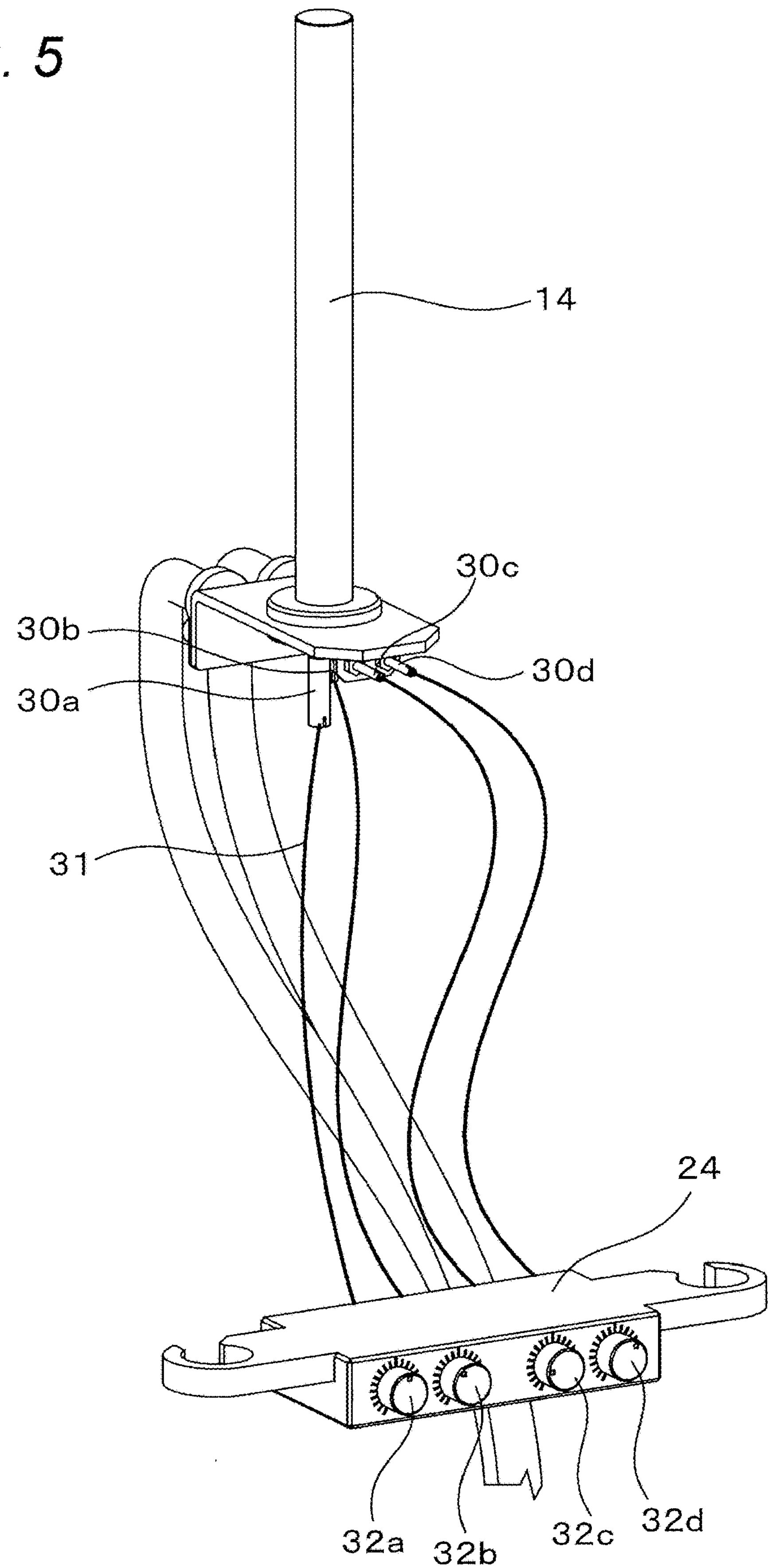


Fig. 6

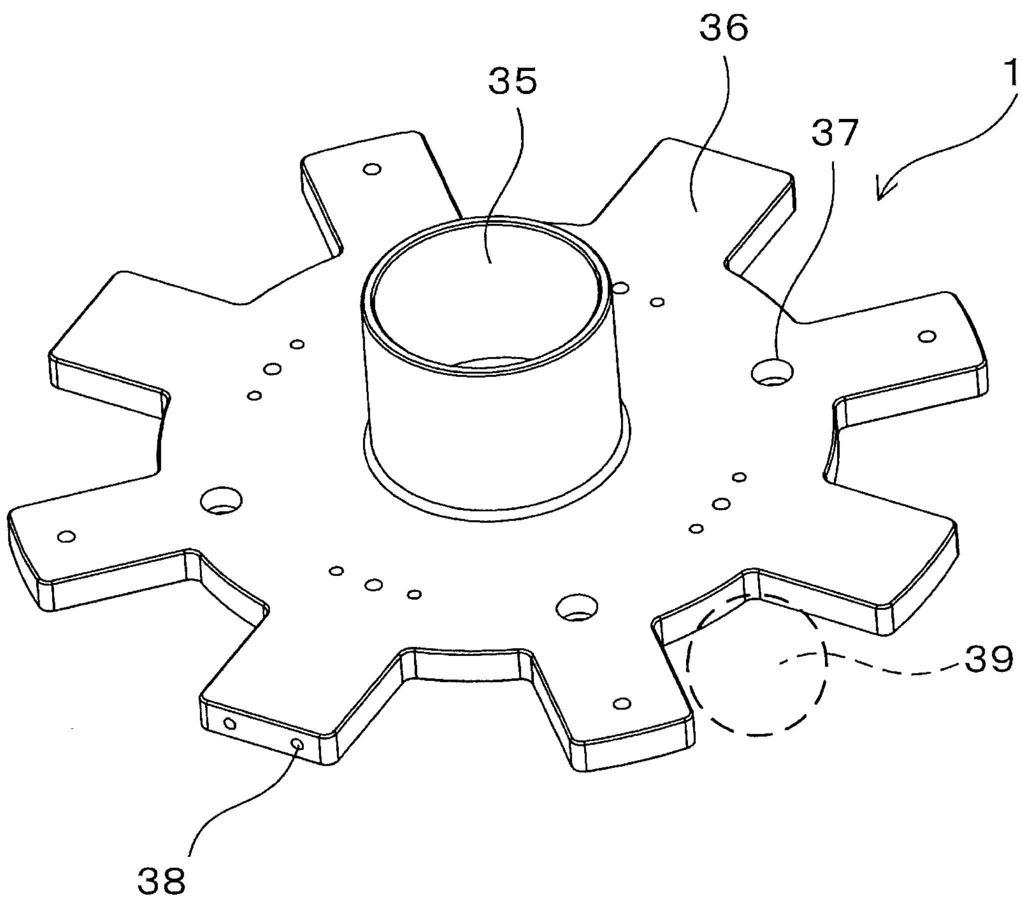


Fig. 7

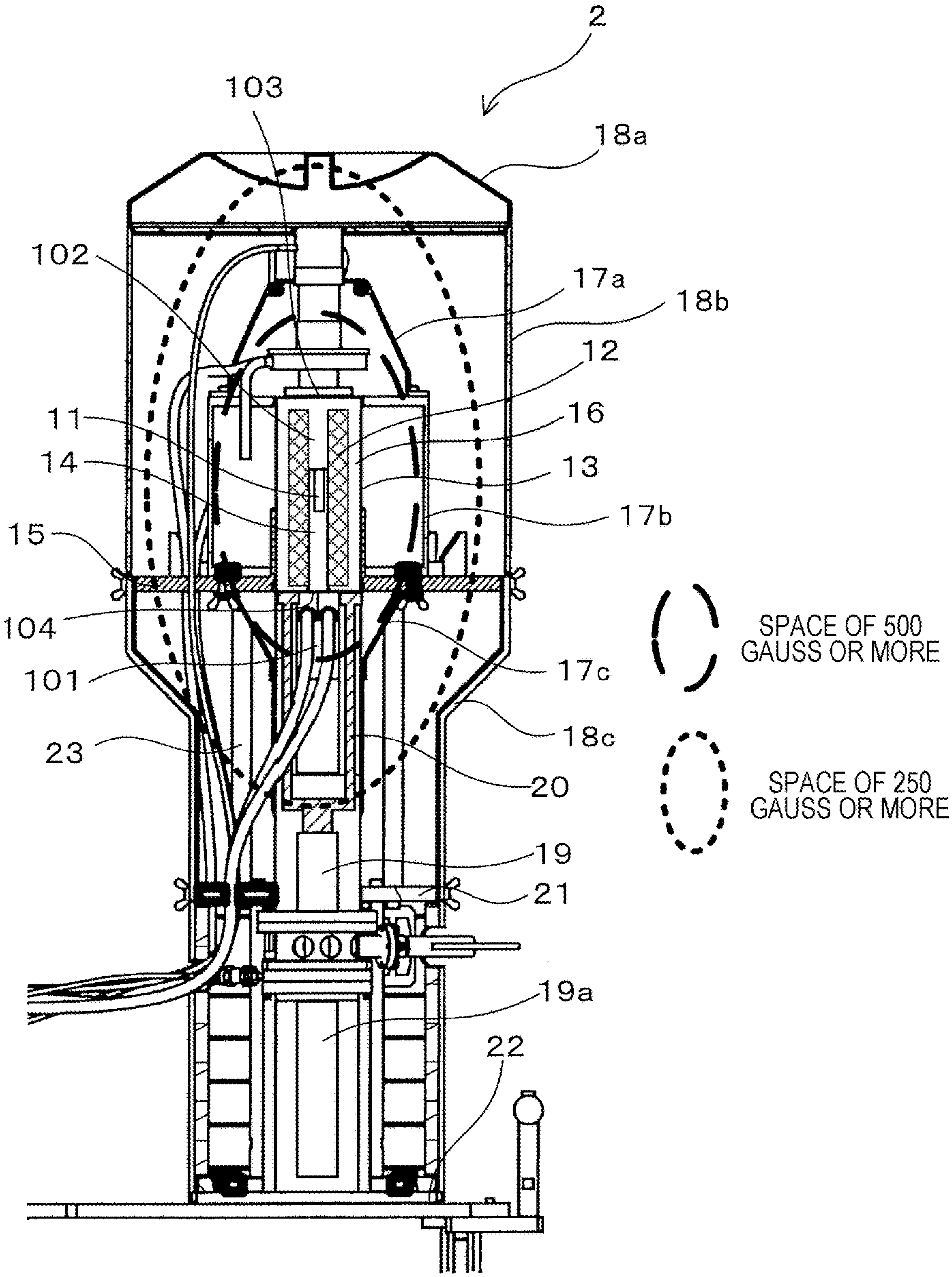

2
103
18a
102
17a
18b
12
11
16
14
13
17b
15
104
17c
101
18c
23
20
19
21
19a
22
SPACE OF 500 GAUSS OR MORE
SPACE OF 250 GAUSS OR MORE

Fig. 8

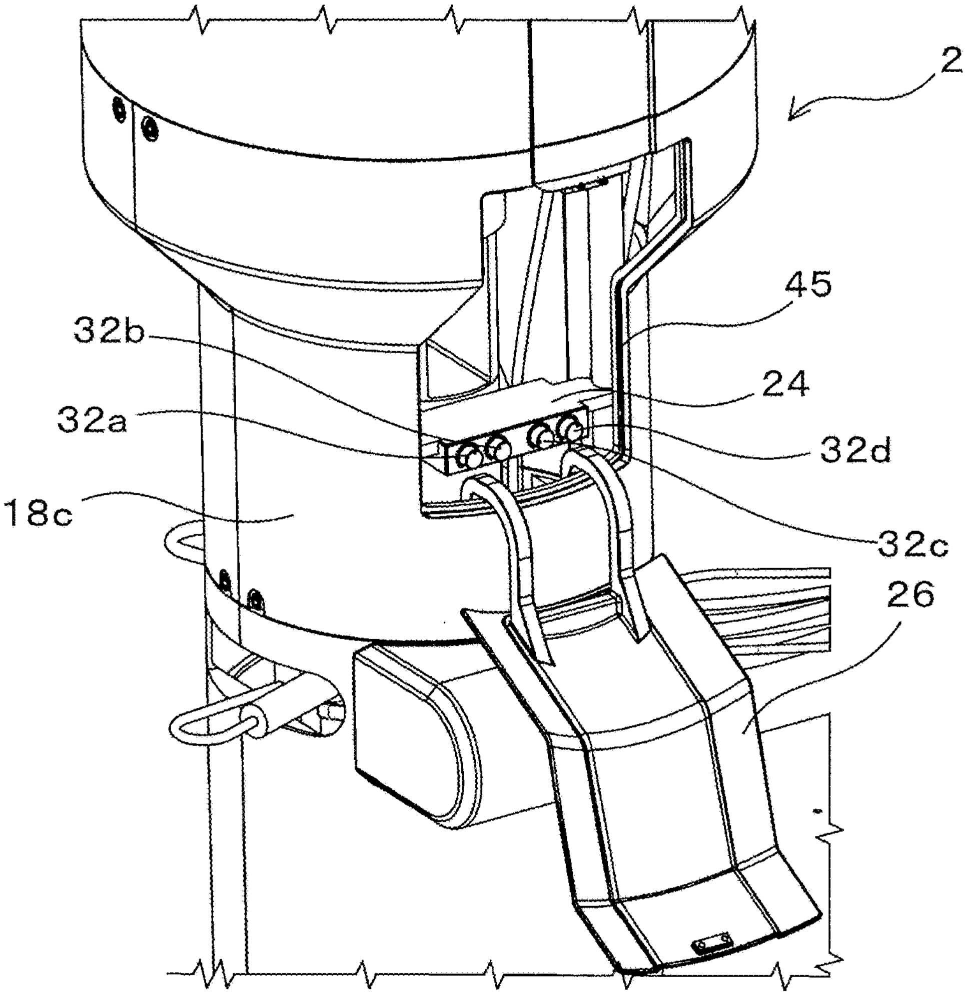

2
45
32b
24
32a
32d
18c
32c
26

Fig. 9

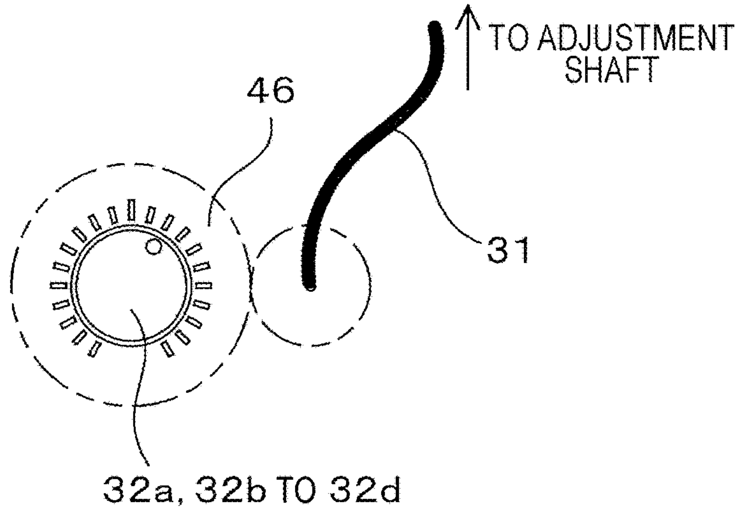
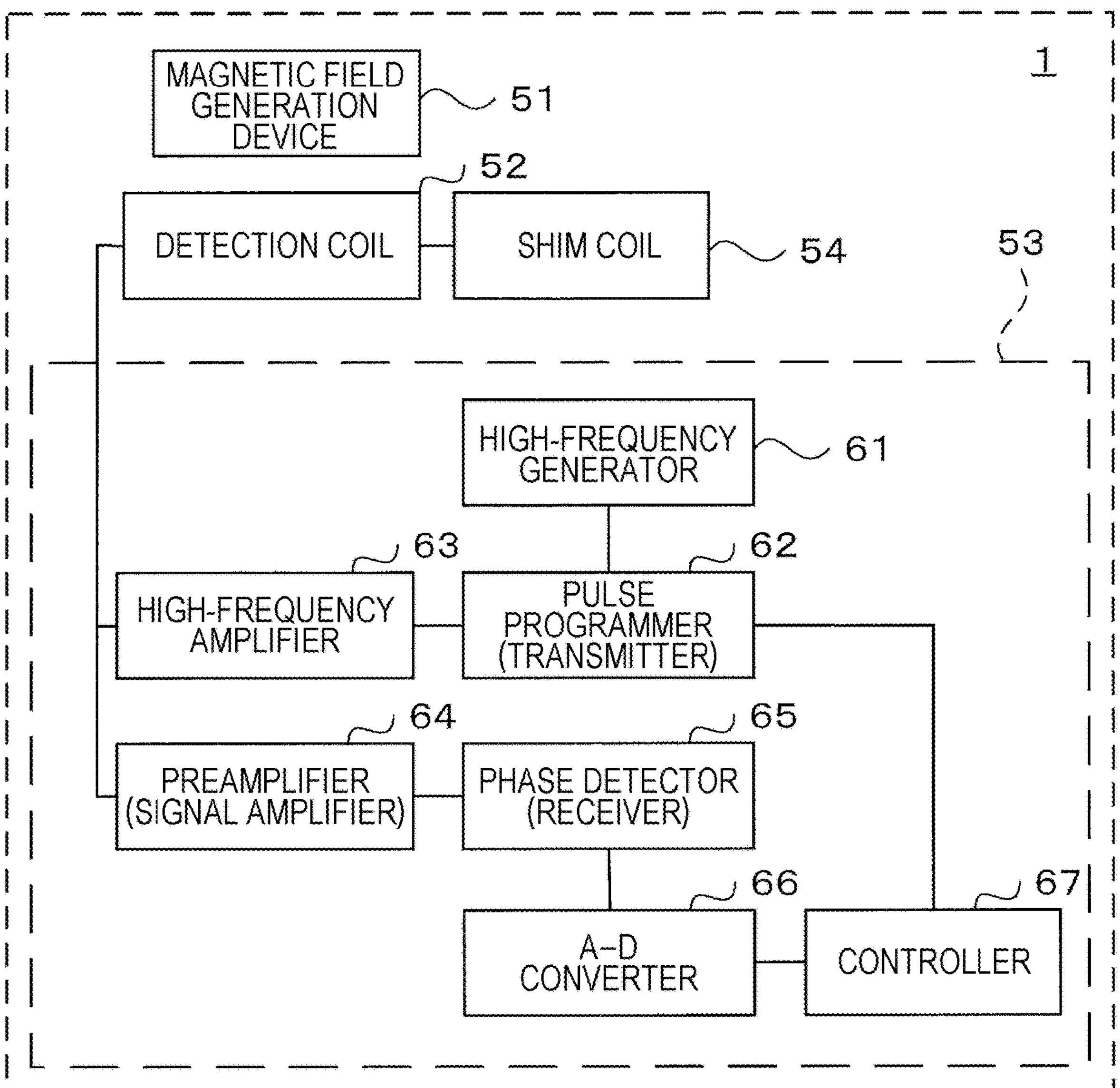

TO ADJUSTMENT SHAFT
46
31
32a, 32b TO 32d

Fig. 10


1
MAGNETIC FIELD GENERATION DEVICE
51
52
DETECTION COIL
SHIM COIL
54
53
HIGH-FREQUENCY GENERATOR
61
63
62
HIGH-FREQUENCY AMPLIFIER
PULSE PROGRAMMER (TRANSMITTER)
64
65
PREAMPLIFIER (SIGNAL AMPLIFIER)
PHASE DETECTOR (RECEIVER)
66
67
A-D CONVERTER
CONTROLLER

NUCLEAR MAGNETIC RESONANCE APPARATUS

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is based on and claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2024-036760, filed on Mar. 11, 2024, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] This disclosure relates to a nuclear magnetic resonance apparatus including a magnetic field generator.

BACKGROUND DISCUSSION

[0003] Conventionally, various analysis techniques and image diagnosis techniques using nuclear magnetic resonance (NMR) have been put into practical use. Nuclear magnetic resonance is a phenomenon in which, when an electromagnetic wave is emitted from outside at an atomic nucleus to which a magnetic field is applied, the atomic nucleus absorbs a specific electromagnetic wave corresponding to each chemical environment. Examples of a nuclear magnetic resonance apparatus utilizing these nuclear magnetic resonance phenomena include a nuclear magnetic resonance analysis apparatus (NMR analysis apparatus) that utilizes nuclear magnetic resonance phenomena to analyze structures of samples, a magnetic resonance imaging apparatus (MRI apparatus) that utilizes nuclear magnetic resonance phenomena to image information of an inside of a living body, and the like.

[0004] These nuclear magnetic resonance apparatuses (hereinafter, abbreviated as an NMR apparatus) internally include a device called a probe as a mechanism for emitting the above-described electromagnetic wave and detecting a nuclear magnetic resonance signal indicating an amount of energy absorbed due to a nuclear magnetic resonance phenomenon. In order to accurately analyze a sample, it is necessary to individually adjust (tune) a characteristic of the probe according to an installed NMR apparatus. Therefore, conventionally, as disclosed in JP 2023-061460 A, a probe includes an operation unit for adjusting characteristics, and an operator adjusts a characteristic of the probe by operating the operation unit.

[0005] Here, as a part thereof, the NMR apparatus includes a magnetic field generator that generates a magnetic field. However, with an NMR apparatus in particular, sensitivity and resolution of a signal are higher as the generated magnetic field is stronger. Therefore, it is desired that the NMR apparatus includes a magnetic field generator that generates a strong magnetic field. However, if such a magnetic field generator that generates a strong magnetic field is provided, as in JP 2023-061460 A (paragraph 0044, FIG. **4**), when an operation unit included in a probe is operated to adjust a characteristic of the probe, a metal object worn or possessed by an operator who operates the operation unit may be attracted, or malfunction of a precision instrument may be caused.

[0006] A need thus exists for a nuclear magnetic resonance apparatus which is not susceptible to the drawback mentioned above.

SUMMARY

[0007] A nuclear magnetic resonance apparatus includes a magnetic field generator that generates a magnetic field, a sample container that is held in a magnetic field generated by the magnetic field generator and contains a sample to be analyzed, a probe that emits an electromagnetic wave at the sample and detects a nuclear magnetic resonance signal from the sample, a first protective cover that surrounds at least a space in which strength of a magnetic field generated by the magnetic field generator is equal to or greater than a first threshold value, a second protective cover that is mounted outside of the first protective cover and surrounds at least a space in which strength of a magnetic field generated by the magnetic field generator is equal to or greater than a second threshold value that is weaker than the first threshold value, an opening provided in the second protective cover, and a probe adjuster that is disposed outside the first protective cover and inside the second protective cover, and in vicinity of the opening, and adjusts a characteristic of the probe by receiving operation by an operator.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The foregoing and additional features and characteristics of this disclosure will become more apparent from the following detailed description considered with the reference to the accompanying drawings, wherein:

[0009] FIG. **1** is an external view showing an entire NMR analysis apparatus according to the present embodiment;

[0010] FIG. **2** is an external view showing the entire NMR analysis apparatus in a state where protective covers covering outside of the NMR analysis apparatus are removed;

[0011] FIG. **3** is a cross-sectional view showing a periphery of a main body of the NMR analysis apparatus in particular;

[0012] FIG. **4** is a view showing a state in which a probe is fitted to a storage container;

[0013] FIG. **5** is a view showing the probe and a probe adjuster;

[0014] FIG. **6** is an external view showing appearance of a stay;

[0015] FIG. **7** is a view showing spaces surrounded by strong-magnetic-field protection members and weak-magnetic-field protection members;

[0016] FIG. **8** is a view showing vicinity of an opening of the main body in a state where an opening and closing door is open;

[0017] FIG. **9** is a diagram showing an example of a dial of the probe adjuster; and

[0018] FIG. **10** is a diagram showing a schematic configuration of the NMR analysis apparatus.

DETAILED DESCRIPTION

[0019] Hereinafter, a nuclear magnetic resonance apparatus according to an embodiment of this disclosure will be described in detail with reference to the drawings. Hereinafter, as the nuclear magnetic resonance apparatus, a nuclear magnetic resonance analysis apparatus (hereinafter, referred to as an NMR analysis apparatus) that analyzes a structure of a sample by using a nuclear magnetic resonance phenomenon in particular will be described as an example.

[0020] First, a schematic configuration of an NMR analysis apparatus **1** according to the present embodiment will be described with reference to FIGS. **1** and **2**. FIG. **1** is an external view showing the entire NMR analysis apparatus **1** according to the present embodiment, and FIG. **2** is an external view showing the entire NMR analysis apparatus **1** in a state where protective covers (weak-magnetic-field protection members **18***a* to **18***c* described later) particularly covering outside of the NMR analysis apparatus **1** are removed.

[0021] As shown in FIGS. **1** and **2**, the NMR analysis apparatus **1** according to the present embodiment basically includes a main body **2** internally including a sample container that contains a sample to be analyzed, a probe for analyzing the sample, and a superconductor (magnetic field generator) for generating a static magnetic field in the main body **2**, an external device **4** connected to the main body **2** through a pipe **3** or a cable and including a compressor that supplies a refrigerant for cooling the superconductor in the main body **2**, a pump for vacuuming the inside of the main body **2**, and the like, and a cart **5** on which the main body **2** and the external device **4** are mounted and integrally movable. However, in the NMR analysis apparatus **1** according to the present embodiment, the cart **5** is not essential, and the main body **2** and the external device **4** may be installed on a floor surface.

[0022] First, the main body **2** of the NMR analysis apparatus **1** will be described below. FIG. **3** is a cross-sectional view showing particularly a periphery of the main body **2** of the NMR analysis apparatus **1**.

[0023] As shown in FIG. **3**, the main body **2** includes a sample container **11** that contains the sample to be analyzed, a superconductor **12** having a cylindrical shape and disposed around the sample container **11**, a storage container **13** having a cylindrical shape as well and housing the superconductor **12**, a probe **14** that emits an electromagnetic wave at the sample in the sample container **11** and detects a nuclear magnetic resonance signal (NMR signal) from the sample, a stay (upper fixing member) **15** having a disk shape and disposed around the storage container **13**, a vacuum thermal insulation container **16** that thermally insulates the superconductor **12** by bringing inside of the storage container **13** into a vacuum state, strong-magnetic-field protection members (first protective covers) **17***a* to **17***c* that are protective covers disposed so as to surround a space of a strong magnetic field in which a strength of a magnetic field generated outside of the storage container **13** by the superconductor **12** is equal to or greater than a first threshold value, weak-magnetic-field protection members (second protective covers) **18***a* to **18***c* that are protective covers disposed so as to surround a space of a weak magnetic field in which a strength of a magnetic field generated outside of the strong-magnetic-field protection members **17***a* to **17***c* by the superconductor **12** is equal to or greater than a second threshold value, a cold head **19** of a refrigerator **19***a* that expands a compressed refrigerant (for example, He gas, Ne gas, H gas, N gas, or gas obtained by liquefying these gases) supplied from the external device **4** to generate cold, a heat transfer body **20** for transferring the cold generated in the cold head **19** of the refrigerator **19***a* to the superconductor **12**, a middle fixing member **21** and lower fixing member **22** having a disk shape, disposed around the storage container **13** and below the stay **15**, and fixing the weak-magnetic-field protection members **18***a* to **18***c* together with the stay **15**, a fixing support **23** for supporting the main body **2**, and a probe adjuster **24** (refer to FIG. **2**) that is attached to the fixing support **23** and adjusts (tunes) a characteristic of the probe **14** by receiving operation by an operator. The stay **15** is held at a predetermined position (height) from a lower end of the storage container **13** by the fixing support **23**. In the present embodiment, the respective strong-magnetic-field protection members **17***a* to **17***c* and weak-magnetic-field protection members **18***a* to **18***c* are divided into three parts. However, the strong-magnetic-field protection members **17***a* to **17***c* may be an integrated protective cover, and the weak-magnetic-field protection members **18***a* to **18***c* may be an integrated protective cover. As shown in FIG. **1**, a window **25** formed by a transmissive member is provided on a side surface of the main body **2**. The window **25** is for the operator to visually recognize inside of the main body **2**. An opening and closing door **26**, which is opened and closed when the probe adjuster **24** is operated as will be described later, is provided below the window **25**. The probe adjuster **24** is attached to outside of a space of 500 gauss or more as shown in FIG. **7**. In the present embodiment, the probe adjuster **24** is attached to outside of the space of 500 gauss or more and inside of the weak-magnetic-field protection members **18***a* to **18***c*, and covered with the opening and closing door **26** so that the operator does not touch the probe adjuster **24** after adjustment. However, this disclosure is not limited to this configuration, and other configurations are possible. As an example, an inexpensive configuration can be considered in which the probe adjuster **24** is embedded in the weak-magnetic-field protection members **18***a* to **18***c*, and dials **32***a* to **32***d* of the probe adjuster **24** are exposed toward outside of the weak-magnetic-field protection members **18***a* to **18***c*.

[0024] Although not described, the main body **2** also includes various devices and members necessary for analysis of a sample by the NMR analysis apparatus **1**.

[0025] Hereinafter, each component included in the main body **2** will be described in order.

[0026] The storage container **13** is, for example, a sealed, substantially cylindrical container. The storage container **13** has a communication space **101** that penetrates the storage container **13** in a radial direction between the weak-magnetic-field protection member **18***a* and the cold head **19** of the refrigerator **19***a*. A room-temperature bore space **102** extending in an axial direction of the storage container **13** is provided around a central axis of the storage container **13**. A first opening **103** directed toward the weak-magnetic-field protection member **18***a* is provided at one end of the room-temperature bore space **102**, and a second opening **104** opened to the communication space **101** is provided at another end of the room-temperature bore space **102**. The room-temperature bore space **102** and the communication space **101** are partitioned and spatially isolated from an internal space of the storage container **13**. By keeping the internal space of the storage container **13** in a vacuum state, the internal space of the storage container **13** is thermally isolated from the room-temperature bore space **102** and the communication space **101**, and the storage container **13** becomes the vacuum thermal insulation container **16**. In a case where the NMR analysis apparatus **1** is used, after a sample of various kinds, such as an organic compound, inorganic compound, and biological substance in various fields such as the pharmaceutical field, the biotechnology field, and the material field, is contained in the sample container **11**, the weak-magnetic-field protection member **18***a* is removed, and the sample container **11** is inserted into the room-temperature bore space **102** of the storage container **13** from the first opening **103** toward the superconductor **12**. Then, the NMR analysis apparatus **1** according to the present embodiment emits a pulsed electromagnetic wave with the probe **14**, on the sample container **11** disposed in a static magnetic field generated by the superconductor **12** to excite nuclear magnetic resonance, and detects a nuclear magnetic resonance signal generated by the nuclear magnetic resonance, thereby analyzing a crystal structure of the sample in the sample container **11**. Note that, although not shown, a monitor for displaying results of these analyses, a storage for storing the analysis results, a keyboard, a mouse, and the like for an operating person to perform operations are also connected to the NMR analysis apparatus **1**.

[0027] Meanwhile, the superconductor **12** corresponds to a magnetic field generation source (magnetic pole) in the NMR analysis apparatus **1**, and generates a static magnetic field for analyzing the sample. As the superconductor **12**, a superconducting bulk magnet, particularly a high-temperature superconducting bulk magnet having a relatively high superconducting transition temperature (critical temperature), is used for a purpose of being able to generate a strong static magnetic field having excellent stability and enabling downsizing of the apparatus. For example, a RE-Ba—Cu—O (RE is a rare-earth element including Y)-based superconductor which is a high-temperature oxide superconductor is desirable. However, the superconductor **12** is not limited to the example described above, and may be an Nd-based, Sm-based, or Gd-based superconducting bulk magnet. In addition, a material other than a high-temperature superconducting bulk magnet may be used as long as it can be maintained at a superconducting transition temperature or lower.

[0028] Then, the superconductor **12** according to the present embodiment is formed in a cylindrical shape so as to surround the sample container **11** as shown in FIG. **3**, and is disposed in the storage container **13** having a cylindrical shape as well. In particular, in the present embodiment, the superconductor **12** is formed by stacking a plurality of (for example, 3 to 10) cylindrical superconducting bulk magnets along the axial direction. However, the superconductor **12** may be formed of one superconducting bulk magnet instead of a plurality of superconductors. As will be described later, the superconductor **12** is thermally connected to the cold head **19** of the refrigerator **19*a*** via the heat transfer body **20**, so that cold generated by the cold head **19** of the refrigerator **19*a*** can be transferred to the superconductor **12**.

[0029] In addition, although the superconducting bulk magnet described above does not need to be energized to maintain a magnetic field, it is necessary to perform the following magnetization process as described in (1) to (3) in advance.

(1) Magnetic Field Application Step

[0030] A magnetic field is applied to a superconductor by an external magnetic field generation device (hereinafter, referred to as a magnetization device) so that, at a temperature higher than a superconducting transition temperature (critical temperature) of the superconductor **12**, a magnetic field in which a magnetic flux passes in the axial direction is generated in an inner peripheral space of the superconductor **12**.

(2) Magnetic Field Cooling Step

[0031] The superconductor **12** to which the magnetic field is applied as described above by the magnetization device is cooled to a temperature TO equal to or lower than the superconducting transition temperature. Specifically, the refrigerator **19*a*** is operated while the magnetic field is applied to the superconductor **12** by operation of the magnetization device. As a result, the refrigerator **19*a*** generates the cold, and the generated cold is transmitted via the cold head **19** to the superconductor **12** thermally connected to the cold head **19**. As a result, the superconductor **12** is cooled to the temperature TO equal to or lower than the superconducting transition temperature. Thereafter, the superconductor **12** is maintained in a cooled state.

(3) Demagnetization Step

[0032] Operation of the magnetization device is stopped while the temperature of the superconductor **12** is maintained at the temperature TO. As a result, the magnetic field applied to the superconductor **12** cooled to a temperature equal to or lower than the superconducting transition temperature (that is, brought into a superconducting state) is removed. Then, a superconducting current is induced in the superconductor **12** so as to restore a state of the magnetic field in response to a change in magnetic field strength associated with the removal of the applied magnetic field. The superconducting current is a circular current flowing in a plane perpendicular to a central axis of the superconductor **12**, around the central axis of the superconductor **12**. The superconducting current induced in this manner flows in the superconductor **12** to generate a magnetic field. That is, the superconductor **12** is magnetized. The magnetic field generated by the magnetization of the superconductor **12** is basically the same magnetic field as the applied magnetic field generated by the operation of the magnetization device. That is, when the superconducting current flows in the superconductor **12**, the superconductor **12** traps the applied magnetic field generated by the operation of the magnetization device. When the superconductor **12** traps the applied magnetic field, a magnetic field (trapped magnetic field) through which a magnetic flux passes in the axial direction is formed in the inner peripheral space of the superconductor **12**. Because the sample container **11** is positioned in the inner peripheral space of the superconductor **12** in which the trapped magnetic field is generated, the sample can be disposed in the static magnetic field. Thus, as described above, it is possible to analyze the sample placed in the strong static magnetic field by applying the electromagnetic wave to the sample from the probe **14**.

[0033] Note that, after the magnetization process in (1) to (3) are performed, it is possible to maintain a state (superconducting characteristic) in which a strong magnetic field is generated for a long time, by maintaining the magnetized superconductor **12** at a temperature equal to or lower than the superconducting transition temperature by using the cold head **19** of the refrigerator **19*a*** and the heat transfer body **20**. In a case where the strong magnetic field is lost thereafter due to a temperature rise or the like, also, it is possible to restore the state in which the strong magnetic field is generated by performing the above-described magnetization process again.

[0034] The storage container **13** has a long cylindrical shape as shown in FIGS. **2** and **3**, and stores the above-described superconductor **12** particularly around the sample container **11** near a top thereof. The heat transfer body **20** to be described later is housed in a central portion of the storage container **13**, and the cold head **19** of the refrigerator **19***a* is housed below the storage container **13**. The stay **15**, the middle fixing member **21**, and the lower fixing member **22** are disposed around the storage container **13**. The refrigerator **19***a* and the cold head **19** are fixed to and supported by the cart **5**. The heat transfer body **20** is supported by and fixed to the cold head **19** of the refrigerator **19***a*. The heat transfer body **20** is provided inside the storage container **13** and at a position surrounding the communication space **101**. The superconductor **12** is supported by and fixed to the heat transfer body **20**. The superconductor **12** is provided inside the storage container **13** and at a position surrounding the room-temperature bore space **102**.

[0035] The probe **14** has a cylindrical shape, is thinner than the room-temperature bore space **102** of the storage container **13**, and is fitted by being inserted through the communication space **101** and the second opening **104** into the room-temperature bore space **102** formed near the central axis of the storage container **13**. As shown in FIG. **3**, the fitted probe **14** is positioned in an internal space of the superconductor **12** having a cylindrical shape, that is, in the static magnetic field generated by the superconductor **12**. FIG. **4** is a view showing the probe **14** fitted in the storage container **13**. FIG. **5** is an extracted view of the probe **14** and the probe adjuster **24** connected to the probe **14**, of the NMR analysis apparatus **1**.

[0036] Here, the probe **14** includes, for example, a high-frequency generator that generates an electromagnetic wave, a coil that emits the electromagnetic wave at the sample container **11** and through which a nuclear magnetic resonance signal (NMR signal) from the sample container **11** flows, a detector that amplifies and detects the nuclear magnetic resonance signal flowing through the coil, a controller that controls these components, and the like. Then, the electromagnetic wave generated by the high-frequency generator is emitted at the sample, and a nuclear magnetic resonance signal from the sample is detected. Thereafter, the detected nuclear magnetic resonance signal converts a signal, which is a time-axis function, into a frequency function by using Fourier transform. Then, the nuclear magnetic resonance signal is finally stored as an NMR spectrum in a computer connected to the outside, and can be displayed on a monitor or the like.

[0037] Meanwhile, a main body of the probe **14** includes adjustment shafts **30***a* to **30***d* for adjusting (tuning) a characteristic of the probe. The adjustment shaft **30***a* is connected to a dial **32***a* of the probe adjuster **24** described later, by a connection mechanism, such as a wire **31** for example, to which rotational force can be applied. As a result, when the operator rotates the dial **32***a*, the rotational force is transmitted to the adjustment shaft **30***a* via the wire **31**, and the adjustment shaft **30***a* can be rotated. Therefore, the operator can remotely operate the adjustment shaft **30***a* by rotating the dial **32***a*. Similarly, the adjustment shaft **30***b* is connected to a dial **32***b* of the probe adjuster **24**, the adjustment shaft **30***c* is connected to a dial **32***c* of the probe adjuster **24**, and the adjustment shaft **30***d* is connected to a dial **32***d* of the probe adjuster **24**.

[0038] The adjustment of the characteristic of the probe **14** by the adjustment shafts **30***a* to **30***d* is, for example, an action of matching a frequency at which an atomic nucleus resonates in the nuclear magnetic resonance phenomenon with a frequency of a pulse output from the probe. Specifically, capacitance to each of a tuning capacitor and a matching capacitor is changed to perform adjustment so that reflection of an irradiation signal at a resonant frequency is minimized. It is also possible to perform drift correction on the superconductor **12** (for example, adjustment of an amplification factor or phase of a lock signal), and sensitivity and resolution of the spectrum are improved by the adjustment. For example, it is assumed that the electric capacitance of the capacitors is adjusted by using the adjustment shaft **30***a* and the adjustment shaft **30***b*, and drift correction is performed by using the adjustment shaft **30***c* and the adjustment shaft **30***d*. If these adjustments are not appropriately performed, a signal-to-noise ratio may be degraded and a required pulse width will increase. Therefore, for example, at a time of initial installation or periodic maintenance of the NMR analysis apparatus **1**, the operator performs adjustment while checking the NMR spectrum displayed on the monitor by using a sample for adjustment or the like. At that time, the operator does not need to directly operate the adjustment shafts **30***a* to **30***d*, and can work by operating the dials **32***a* to **32***d* of the probe adjuster **24** at a distant position.

[0039] Here, as will be described later, the probe **14** is in the space of the strong magnetic field where the strength of the magnetic field is equal to or greater than the first threshold value, but the probe adjuster **24** is in the space of the weak magnetic field where the magnetic field is less than the first threshold value and equal to or greater than the second threshold value, which is weaker than the first threshold value. Therefore, it is possible to prevent a metal object worn or possessed by the operator from being attracted, and malfunction of a precision instrument.

[0040] Meanwhile, the stay **15** has a disk shape and is formed of a non-magnetic material (for example, a resin material, titanium, aluminum, or the like), and is disposed around the storage container **13**. A large number of screw holes and the like are formed in the stay **15**, and correspond to fixing members used for fixing the strong-magnetic-field protection members **17***a* to **17***c* and the weak-magnetic-field protection members **18***a* to **18***c*, which will be described later, to the main body **2**.

[0041] As shown in FIG. **6**, the stay **15** includes a cylindrical portion **35** that surrounds an outer wall and outer periphery of the storage container **13**, and flanges **36** extending in the radial direction from a lower end of the cylindrical portion **35**. The cylindrical portion **35** has a cylindrical shape corresponding to the cylindrical shape of the storage container **13**, and is disposed at a predetermined position (height) from the lower end of the storage container **13** so as to surround the outer wall and outer periphery of the storage container **13**. An inner diameter of the cylindrical portion **35** is set to a length corresponding to an outer diameter of the storage container **13** (the inner diameter of the cylindrical portion **35** is slightly larger than the outer diameter of the storage container **13**, and has a length that allows the storage container **13** to pass through inside of the cylindrical portion **35** and with which a relative position can be fixed unless load is applied). The stay **15** and the storage container **13** are fitted to each other by, for example, interference fit at the cylindrical portion **35**. However, before the fitting, the stay **15** is movable in the axial direction with respect to the storage container **13** (a position of the stay **15** in a height direction can be adjusted) by applying load. The position of the stay **15** (position from the lower end of the storage container **13**) is finally determined by the operator fitting the fixing support **23**.

[0042] As shown in FIG. **6**, a large number of screw holes **37** are formed in the flanges **36** in the axial direction (vertical direction). Screw holes **38** are formed in outer peripheral surfaces of the flanges **36** in the radial direction. Here, as shown in FIG. **3**, the screw holes **37** in the axial direction are used to fix the strong-magnetic-field protection members **17***a* to **17***c*. Specifically, the strong-magnetic-field protection member **17***a* and the strong-magnetic-field protection member **17***b* are fixed to an upper side of the stay **15** by screws (however, the strong-magnetic-field protection member **17***a* is not directly fixed to the stay **15**, but is fixed via the strong-magnetic-field protection member **17***b*), and the strong-magnetic-field protection member **17***c* is fixed to a lower side of the stay **15** by screws.

[0043] As shown in FIG. **6**, the screw holes **38** in the radial direction are used to fix the weak-magnetic-field protection members **18***a* to **18***c*. Specifically, the weak-magnetic-field protection member **18***a* and the weak-magnetic-field protection member **18***b* are fixed to the upper side of the stay **15** by screws (however, the weak-magnetic-field protection member **18***a* is not directly fixed to the stay **15**, but is fitted as a lid to a top of the weak-magnetic-field protection member **18***b*), and the weak-magnetic-field protection member **18***c* is fixed to the lower side of the stay **15** by screws. The weak-magnetic-field protection member **18***c* is also fixed by screws to the middle fixing member **21** and lower fixing member **22** positioned below the stay **15**.

[0044] Meanwhile, as shown in FIG. **6**, as viewed from above, the flanges **36** form a gear shape having a plurality of gaps **39** between teeth. The gaps **39** may be passages of cables and pipes running between the upper side and lower side of the stay **15** inside the main body **2**. Because there are the plurality of gaps **39** in the radial direction, gaps **39** through which the cables and pipes that run can be freely selected according to positions of the cables and pipes. As a result, as shown in FIGS. **2** and **3**, the cables and pipes connected to an upper portion of the storage container **13** can be taken out from the lower end of the main body **2**. Note that a shape other than the gear shape may be applied as long as the shape has gaps **39** through which cables and pipes can pass.

[0045] Meanwhile, the vacuum thermal insulation container **16** is disposed inside the storage container **13**. The above-described superconductor **12** is disposed further inside the vacuum thermal insulation container **16**. The vacuum thermal insulation container **16** has an internal space, and the internal space is in a vacuum state. By bringing the internal space into a vacuum state, it is possible to thermally insulate the superconductor **12** disposed inside.

[0046] The strong-magnetic-field protection members **17***a* to **17***c* are formed of a non-magnetic material (for example, a resin material, titanium, aluminum, or the like), and are protective covers disposed so as to surround the space of the strong magnetic field in which the strength of the magnetic field generated outside the storage container **13** by the superconductor **12** is equal to or greater than the first threshold value. Here, the first threshold value is, for example, 500 gauss. The space of the strong magnetic field in which the strength of the magnetic field is equal to or greater than the first threshold value corresponds to a space that requires prohibition of any magnetic material approaching the superconductor **12**. As shown in FIG. **7**, the strong-magnetic-field protection members **17***a* to **17***c* surround the space of the strong magnetic field in which the strength of the magnetic field is equal to or greater than the first threshold value, by which it is possible to reliably prevent entry of a metal object worn or possessed by the operator into the area. Note that the strong-magnetic-field protection members **17***a* to **17***c* may surround an area larger than the space of the strong magnetic field in which the strength of the magnetic field generated by the superconductor **12** is equal to or greater than the first threshold value, as long as the strong-magnetic-field protection members **17***a* to **17***c* can at least surround the space of the strong magnetic field. However, for a purpose of reducing a size of the apparatus, the size is preferably as small as possible while satisfying a condition that the strong-magnetic-field protection members **17***a* to **17***c* surround the space of the strong magnetic field in which the strength of the magnetic field is equal to or greater than the first threshold value.

[0047] The weak-magnetic-field protection members **18***a* to **18***c* are formed of a non-magnetic material (for example, a resin material, titanium, aluminum, or the like) as well, and are protective covers disposed so as to surround the space of the weak magnetic field in which the strength of the magnetic field generated outside of the strong-magnetic-field protection members **17***a* to **17***c* by the superconductor **12** is equal to or greater than the second threshold value. Here, the second threshold value is, for example, 250 gauss. The space of the weak magnetic field in which the strength of the magnetic field is equal to or greater than the second threshold value corresponds to a space that requires prohibition of any precision instrument approaching the superconductor **12**. As shown in FIG. **7**, the weak-magnetic-field protection members **18***a* to **18***c* surround the space of the weak magnetic field in which the strength of the magnetic field is equal to or greater than the second threshold value, by which it is possible to reliably prevent entry of precision instrument worn or possessed by the operator into the area. Note that the weak-magnetic-field protection members **18***a* to **18***c* may surround an area larger than the space of the weak magnetic field in which the strength of the magnetic field generated by the superconductor **12** is equal to or greater than the second threshold value, as long as the weak-magnetic-field protection members **18***a* to **18***c* can at least surround the space of the weak magnetic field. However, for a purpose of reducing a size of the apparatus, the size is preferably as small as possible while satisfying a condition that the weak-magnetic-field protection members **18***a* to **18***c* surround the space of the weak magnetic field in which the strength of the magnetic field is equal to or greater than the second threshold value. As a result, in the present embodiment, as shown in FIG. **1**, the main body **2** has a larger diameter on an upper side thereof than on a lower side thereof (has an inverted triangle shape).

[0048] In the present embodiment, after the superconductor **12** is magnetized and the strong-magnetic-field protection members **17***a* to **17***c* and the weak-magnetic-field protection members **18***a* to **18***c* are fitted to the stay **15**, the strong-magnetic-field protection members **17***a* to **17***c* and the weak-magnetic-field protection members **18***a* to **18***c* basically remain mounted all the time until it is necessary to magnetize the superconductor **12** again (however, the weak-magnetic-field protection member **18***a* may be removed when a sample is set into the sample container **11**). Therefore, it is possible to prevent entry of a metal object or precision instrument worn or possessed by the operator, from every direction into an area of a magnetic field having a strength equal to or greater than the first threshold value, or equal to or greater than the second threshold value not only at a time of use of the NMR analysis apparatus **1** needless to say, but also until work of exterior assembly at a time of transportation or installation, periodic maintenance, storage, or the like when the NMR analysis apparatus **1** is not in use is completed.

[0049] The probe **14** is positioned inside the strong-magnetic-field protection members **17***a* to **17***c*, that is, in the space of the strong magnetic field in which the strength of the magnetic field is equal to or greater than the first threshold value. Meanwhile, the probe adjuster **24** is positioned outside the strong-magnetic-field protection members **17***a* to **17***c* where the magnetic field is weaker and inside the weak-magnetic-field protection members **18***a* to **18***c*, that is, in the space of the weak magnetic field in which the strength of the magnetic field is less than the first threshold value and equal to or greater than the second threshold value. Therefore, in a case of adjusting (tuning) the characteristic of the probe **14** as described above, the adjustment can be performed by operation of the probe adjuster **24**. Therefore, it is possible to prevent entry of a metal object or precision instrument worn or possessed by the operator, from every direction into at least an area of a magnetic field having a strength equal to or greater than the first threshold value until work is completed.

[0050] Meanwhile, the refrigerator **19***a* is a device that is airtightly fixed below the storage container **13** and expands a compressed refrigerant (for example, He gas, Ne gas, H gas, N gas, or gas obtained by liquefying these gases) supplied from the external device **4** to cause the cold head **19** stored in the storage container **13** to generate cold. A type of the refrigerator is not limited, but the refrigerator is desirably as small as possible, and is a device capable of generating cold for cooling at least the superconductor **12** to a superconducting transition temperature (critical temperature) or lower. Examples of the refrigerator include a GM refrigerator, a Stirling refrigerator, a pulse tube refrigerator, and the like. As an example, the pulse tube refrigerator is a refrigerator that generates cold by generating a periodic pressure vibration in a pulse tube and expanding and contracting a so-called gas piston in the pulse tube. As long as the refrigerator can generate cold necessary for the cold head **19**, a refrigerator other than the refrigerator according to the present embodiment can be used.

[0051] Meanwhile, the heat transfer body **20** has one end thermally connected to the cold head **19** of the refrigerator **19***a* and another end thermally connected to the superconductor **12**, and transfers the cold generated in the cold head **19** of the refrigerator **19***a* to the superconductor **12**. The heat transfer body **20** is formed of a non-magnetic material (for example, copper) having high thermal conductivity. A vacuum state is maintained around the cold head **19** of the refrigerator **19***a* and the heat transfer body **20** for thermal insulation.

[0052] The probe adjuster **24** is an operation mechanism that adjusts the characteristic of the probe **14** by receiving operation by the operator, and, as shown in FIG. **2**, is fixed to the fixing support **23**, outside the strong-magnetic-field protection members **17***a* to **17***c* and inside the weak-magnetic-field protection members **18***a* to **18***c* as described above. A position at which the probe adjuster **24** is fixed is particularly in vicinity of the opening and closing door **26** (that is, an opening **45** opened by opening the opening and closing door **26**) formed in the weak-magnetic-field protection member **18***c*, particularly at a position facing (corresponding to) the opening **45** at the same height as the opening **45**. FIG. **8** is a view showing a periphery of opening **45** of main body **2** in a state where the opening and closing door **26** is opened.

[0053] As shown in FIG. **8**, the probe adjuster **24** is at a position facing particularly the opening **45**, that is, at a position close to the opening **45** when the opening and closing door **26** is opened. Therefore, the operator can visually recognize the dials **32***a* to **32***d* from the opening **45** and can easily operate the dials **32***a* to **32***d* from the opening **45** by hand. For example, adjustment can be easily performed while checking the NMR spectrum displayed on the monitor.

[0054] As shown in FIG. **4**, scales each indicating a reference position (0 position) for each of the dials **32***a* to **32***d* and an amount of operation from each of the reference positions are drawn on the probe adjuster **24**, and the characteristic of the probe is adjusted according to the operation amount of the dials from the reference position. Content of the adjustment can be clearly specified by a numerical value of the scales. For example, if the adjusted content is recorded as “2” to the right, “3” to the left, or the like, the adjustment content can be shared among a plurality of operators, making readjustment work easier. In the example shown in FIG. **4**, the 12:00 direction is the reference position. Note that, for easy visual recognition of the operation amount, a display may be provided for displaying the reference positions for the dials **32***a* to **32***d* and amount of operation from the reference positions in numerals (−5 to 0 to +5, for example).

[0055] As described above, the adjustment shafts **30***a* to **30***d* are connected to the dials **32***a* to **32***d* on the probe adjuster **24** via the wires **31**, and when the operator rotates the dials **32***a* to **32***d*, rotational force thereof is transmitted to the adjustment shafts **30***a* to **30***d* via the wires **31** by which the adjustment shafts **30***a* to **30***d* can be rotated. Rotation angles of the dials **32***a* to **32***d* on the probe adjuster **24** and rotation angles of the adjustment shafts **30***a* to **30***d* may or may not match with each other. As a case of not matching, as shown in FIG. **9**, a step-up device **46** (or may be a reduction device) that changes a rotation ratio may be incorporated into each of the dials **32***a* to **32***d*. Thus, for example, the adjustment shafts **30***a* to **30***d* are rotated by an angle greater or smaller than a predetermined angle by rotating the dials **32***a* to **32***d* by the predetermined angle. By adjusting a gear ratio of the step-up device **46**, rotation amounts of the adjustment shafts **30***a* to **30***d* with respect to the dials **32***a* to **32***d* can be appropriately adjusted. As a result, for example, even in a specification in which a variable capacitor, a variable resistor, or the like integrated with the adjustment shafts **30***a* to **30***d* requires an adjustment margin of one rotation or more, the dials **32***a* to **32***d* can be adjusted by one rotation or less.

[0056] The window **25** is disposed on a side surface of the main body, and is formed of, for example, a transmissive member in order for the operator to visually recognize inside of the main body **2**. The operator can check an internal state of the main body **2** through the window **25**.

[0057] The opening and closing door **26** includes, for example, a hinge or the like, and has a structure openable and closable by the operator. As shown in FIG. **8**, the characteristic of the probe **14** can be adjusted from the opening **45** by opening the opening and closing door **26**. Therefore, it is not necessary to remove the weak-magnetic-field protection members **18***a* to **18***c* for adjustment of the probe **14**. The opening and closing door **26** is closed and sealed off with a seal or the like, except when the characteristic of the probe **14** is adjusted. As a result, it is possible to reduce chances to come into contact with the probe adjuster **24** except for in a situation where the characteristic of the probe **14** is adjusted, and to prevent an adjusted state from being lost due to erroneous operation. In the configuration described above (that is the configuration in which the probe adjuster **24** is embedded in the weak-magnetic-field protection members **18***a* to **18***c*, and the dials **32***a* to **32***d* of the probe adjuster **24** are exposed toward outside of the weak-magnetic-field protection members **18***a* to **18***c*), it is also conceivable to cover the entire dials **32***a* to **32***d* with a separate cover after adjustment of the dials **32***a* to **32***d*, in order to prevent erroneous operation by the operator.

[0058] Next, with reference back to FIG. **1**, the external device **4** of the NMR analysis apparatus **1** will be described. The external device **4** is connected to the main body **2** described above through the pipe **3** or the cable, and includes the compressor that supplies the refrigerant for cooling the superconductor **12**, the pump for vacuuming the inside of the main body **2**, and the like, Note that, for example, a compressed refrigerant (for example, He gas, Ne gas, H gas, N gas, or gas obtained by liquefying these gases) is supplied from the external device **4**, and the supplied refrigerant is expanded to generate cold in the refrigerator **19***a* of the main body **2**. Note that, although the main body **2** and the external device **4** are separated from each other in the present embodiment, the external device **4** may be disposed in the main body **2**.

[0059] Finally, the cart **5** of the NMR analysis apparatus **1** will be described with reference to FIG. **1**. The cart **5** includes a pedestal **41** on which the main body **2** and the external device **4** are placed, wheels **42** attached to a bottom surface of the pedestal **41**, and a handle **43** gripped by the operator. The operator can easily move the main body **2** and the external device **4** to any positions, for example, by gripping the handle **43** and applying the load. In particular, because the NMR analysis apparatus **1** of the present embodiment is smaller and lighter than a conventional apparatus, the NMR analysis apparatus **1** can be easily moved even by human power. The main body **2** and the external device **4** are fixed to the pedestal **41** by a fixing mechanism (for example, bolts or the like) (not shown), and the main body **2** and the external device **4** are basically operated on the cart **5**, and are stored while being placed on the cart **5** even after the operation is finished. The wheels **42** include a lock mechanism also, and can fix the position of the NMR analysis apparatus **1** so that the NMR analysis apparatus **1** does not move during operation.

[0060] Next, a control configuration of the NMR analysis apparatus **1** having the above-described configuration will be briefly described. FIG. **10** is a diagram showing a schematic configuration of the NMR analysis apparatus **1**.

[0061] The NMR analysis apparatus **1** includes a magnetic field generation device **51** for generating a magnetic field including the superconductor **12** and the refrigerator **19***a* described above, a detection coil **52**, and an analysis mechanism **53**. The detection coil **52** is disposed in the inner peripheral space of the superconductor **12**, and the sample to be analyzed is placed on the inner peripheral side of the detection coil **52**. A shim coil **54** is disposed on an outer peripheral side of the detection coil **52**. The analysis mechanism **53** further includes a high-frequency generator **61**, a pulse programmer (transmitter) **62**, a high-frequency amplifier **63**, a preamplifier (signal amplifier) **64**, a phase detector (receiver) **65**, an analog-digital (A-D) converter **66**, and a controller **67**.

[0062] In a state where the magnetic field generation device **51** is operated, a trapped magnetic field is formed in the inner peripheral space of the superconductor **12**, as described above. Next, by adjusting the trapped magnetic field by the shim coil **54**, uniformity of the magnetic field strength of the trapped magnetic field is improved. At this time, the trapped magnetic field is adjusted by the shim coil **54** so that the uniformity of the strength of the magnetic field in the space is 1 ppm or less. Then, after the uniformity of the strength of the magnetic field is improved, the sample is placed in the sample container **11** in the inner peripheral space of the superconductor **12**. In this state, the high-frequency generator **61** is operated. Then, high-frequency pulses generated by the high-frequency generator **61** are sent to the detection coil **52** via the pulse programmer **62** and the high-frequency amplifier **63**, and a pulse electromagnetic wave is emitted at the sample. By nuclear magnetic resonance that occurs when an electromagnetic wave is emitted at a sample placed in a magnetic field, a very small current flows through the detection coil **52** positioned around the sample. A signal (NMR signal) representing the very small current is transmitted to the controller **67** via the preamplifier **64**, the phase detector **65**, and the A-D converter **66**. The controller **67** calculates an NMR spectrum on the basis of the transmitted NMR signal. A molecular structure of the sample is analyzed on the basis of the obtained NMR spectrum.

[0063] As described above in detail, the NMR analysis apparatus **1** according to the present embodiment includes a superconductor **12** that generates a magnetic field, a sample container **11** that is held in a magnetic field generated by the superconductor **12** and contains a sample to be analyzed, a probe **14** that emits an electromagnetic wave at the sample and detects a nuclear magnetic resonance signal from the sample, strong-magnetic-field protection members **17***a* to **17***c* that surround at least a space in which strength of a magnetic field generated by the superconductor **12** is equal to or greater than a first threshold value, weak-magnetic-field protection members **18***a* to **18***c* that are mounted outside of the strong-magnetic-field protection members **17***a* to **17***c* and surround at least a space in which strength of a magnetic field generated by the superconductor **12** is equal to or greater than a second threshold value that is weaker than the first threshold value, an opening **45** provided in the weak-magnetic-field protection members **18***a* to **18***c*, and a probe adjuster **24** that is disposed outside the strong-magnetic-field protection members **17***a* to **17***c* and inside the weak-magnetic-field protection members **18***a* to **18***c*, and in vicinity of the opening **45**, and adjusts a characteristic of the probe **14** by receiving operation by an operator. Therefore, when adjusting the characteristic of the probe **14**, it is possible to reliably prevent entry of a metal object or precision instrument worn or possessed by the operator into an area in which a strong magnetic field is generated. Meanwhile, because the weak-magnetic-field protection members **18***a* to **18***c* include the probe adjuster **24** therein, it is possible to reduce chances to come into contact with the probe adjuster **24** except for in a situation where the characteristic of the probe **14** is adjusted, and to prevent an adjusted state from being lost due to erroneous operation.

[0064] The probe adjuster **24** includes at least one operation unit (dials **32***a* to **32***d*) that is able to be operated by the operator, the characteristic of the probe **14** is adjusted by the operation unit being operated, and the probe adjuster **24** is disposed in a manner that the operation unit and the opening **45** are positioned to face each other. Therefore, the operator can easily operate the operation unit from the opening **45**. It is also easy for the operator to perform adjustment while checking the NMR spectrum displayed on the monitor.

[0065] The scale indicating a reference position for the operation unit and an amount of operation from the reference position is drawn on the probe adjuster **24**, and the characteristic of the probe **14** is adjusted according to the amount of operation from the reference position for the operation unit. Therefore, content of the adjustment can be clearly specified by a numerical value of the scale. For example, if the adjusted content is recorded as "2" to the right, "3" to the left, or the like, the adjustment content can be shared among a plurality of operators, making readjustment work easier.

[0066] In addition, the weak-magnetic-field protection members **18***a* to **18***c* include the opening and closing door **26** that opens and closes the opening **45**. Therefore, it is possible to easily perform specific adjustment of the probe when the opening and closing door **26** is open. Meanwhile, it is possible to reduce chances to come into contact with the probe adjuster **24** when the opening and closing door **26** is closed, that is, except for in a situation where the characteristic of the probe **14** is adjusted, and to prevent an adjusted state from being lost due to erroneous operation.

[0067] Note that this disclosure is not limited to the above-described embodiments, and various improvements and modifications may be made without departing from the gist of this disclosure.

[0068] For example, in the present embodiment, the strong-magnetic-field protection members **17***a* to **17***c* surround the space of a strong magnetic field of 500 gauss or more around the superconductor **12**, and the weak-magnetic-field protection members **18***a* to **18***c* surround the space of a weak magnetic field of 250 gauss or more around the superconductor **12**. However, the strength of the magnetic field in the space to be surrounded can be appropriately changed. Shapes of the strong-magnetic-field protection members **17***a* to **17***c* and the weak-magnetic-field protection members **18***a* to **18***c* are not limited to the shapes shown in FIGS. **1** and **2**.

[0069] In the present embodiment, the probe adjuster **24** includes the four dials **32***a* to **32***d* as the operation unit. The number of the dials **32***a* to **32***d* corresponds to the number of the adjustment shafts **30***a* to **30***d* included in the probe **14**. Therefore, for example, if the number of the adjustment shafts included in the probe **14** is one, the number of the dials is also one.

[0070] In the present embodiment, the adjustment shafts **30***a* to **30***d* included in the probe **14** and the dials **32***a* to **32***d* included in the probe adjuster **24** are connected to each other by the wires **31**, but may be connected by a mechanism other than wires, as long as the mechanism can transmit rotational force. For example, it is also possible to use a shaft connected by one or more universal joints. Alternatively, the adjustment shafts **30***a* to **30***d* and the dials **32***a* to **32***d* may be electrically connected to each other. For example, actuators driven according to operation of the dials **32***a* to **32***d* may be attached to the adjustment shafts **30***a* to **30***d*.

[0071] Furthermore, in the present embodiment, a nuclear magnetic resonance analysis apparatus that analyzes a structure of a sample by using a nuclear magnetic resonance phenomenon in particular has been described as an example of a nuclear magnetic resonance apparatus including a magnetic field generator. However, this disclosure can also be applied to a magnetic resonance imaging apparatus (MRI apparatus) that images information of an inside of a living body by using a nuclear magnetic resonance phenomenon.

[0072] A nuclear magnetic resonance apparatus includes a magnetic field generator that generates a magnetic field, a sample container that is held in a magnetic field generated by the magnetic field generator and contains a sample to be analyzed, a probe that emits an electromagnetic wave at the sample and detects a nuclear magnetic resonance signal from the sample, a first protective cover that surrounds at least a space in which strength of a magnetic field generated by the magnetic field generator is equal to or greater than a first threshold value, a second protective cover that is mounted outside of the first protective cover and surrounds at least a space in which strength of a magnetic field generated by the magnetic field generator is equal to or greater than a second threshold value that is weaker than the first threshold value, an opening provided in the second protective cover, and a probe adjuster that is disposed outside the first protective cover and inside the second protective cover, and in vicinity of the opening, and adjusts a characteristic of the probe by receiving operation by an operator.

[0073] The nuclear magnetic resonance apparatus having the above-described configuration according to this disclosure includes the first protective cover and the second protective cover around the magnetic field generator, and includes, outside the first protective cover, the probe adjuster for adjusting the characteristic of the probe. Therefore, when adjusting the characteristic of the probe, it is possible to reliably prevent entry of a metal object or precision instrument worn or possessed by the operator into an area in which a strong magnetic field is generated. Meanwhile, because the second protective cover includes the probe adjuster therein, it is possible to reduce chances to come into contact with the probe adjuster except for in a situation where the characteristic of the probe is adjusted, and to prevent an adjusted state from being lost due to erroneous operation.

[0074] In the nuclear magnetic resonance apparatus, the probe adjuster includes at least one operation unit that is able to be operated by the operator, the characteristic of the probe is adjusted by the operation unit being operated, and the probe adjuster is disposed in a manner that the operation unit and the opening are positioned to face each other.

[0075] According to the nuclear magnetic resonance apparatus having the above-described configuration according to this disclosure, the operator can easily operate the operation unit from the opening.

[0076] In the nuclear magnetic resonance apparatus, a scale indicating a reference position for the operation unit and an amount of operation from the reference position is drawn on the probe adjuster, and the characteristic of the probe is adjusted according to the amount of operation from the reference position for the operation unit.

[0077] According to the nuclear magnetic resonance apparatus having the above-described configuration according to this disclosure, content of the adjustment can be clearly specified by a numerical value of the scale. For example, if the adjusted content is recorded as "2" to the right, "3" to the left, or the like, the adjustment content can be shared among a plurality of operators, making readjustment work easier.

[0078] In the nuclear magnetic resonance apparatus, the second protective cover includes an opening and closing door that opens and closes the opening.

[0079] According to the nuclear magnetic resonance apparatus having the above-described configuration according to this disclosure, it is possible to easily perform specific adjustment of the probe when the opening and closing door is open. Meanwhile, it is possible to reduce chances to come into contact with the probe adjuster when the opening and closing door is closed, that is, except for in a situation where the characteristic of the probe is adjusted, and to prevent an adjusted state from being lost due to erroneous operation.

[0080] In the nuclear magnetic resonance apparatus, the sample container is inserted into the magnetic field from one side of the magnetic field generator, and the probe is inserted into the magnetic field from another side of the magnetic field generator.

[0081] In the nuclear magnetic resonance apparatus, the second protective cover includes a removable protection member on one side of the magnetic field generator, and the sample container is inserted into the magnetic field from one side of the magnetic field generator after the removable protection member is removed.

[0082] In the nuclear magnetic resonance apparatus, the second protective cover includes the opening on another side of the magnetic field generator.

[0083] The principles, preferred embodiment and mode of operation of the present invention have been described in the foregoing specification. However, the invention which is intended to be protected is not to be construed as limited to the particular embodiments disclosed. Further, the embodiments described herein are to be regarded as illustrative rather than restrictive. Variations and changes may be made by others, and equivalents employed, without departing from the spirit of the present invention. Accordingly, it is expressly intended that all such variations, changes and equivalents which fall within the spirit and scope of the present invention as defined in the claims, be embraced thereby.

1. A nuclear magnetic resonance apparatus comprising:

a magnetic field generator configured to generate a magnetic field;

a sample container that is held in a magnetic field generated by the magnetic field generator and configured to contain a sample to be analyzed;

a probe configured to emit an electromagnetic wave at the sample and detect a nuclear magnetic resonance signal from the sample;

a first protective cover that surrounds at least a space in which strength of a magnetic field generated by the magnetic field generator is equal to or greater than a first threshold value;

a second protective cover that is mounted outside of the first protective cover and surrounds at least a space in which strength of a magnetic field generated by the magnetic field generator is equal to or greater than a second threshold value that is weaker than the first threshold value;

an opening provided in the second protective cover; and

a probe adjuster that is disposed outside the first protective cover and inside the second protective cover, and in vicinity of the opening, and configured to adjust a characteristic of the probe by receiving operation by an operator.

2. The nuclear magnetic resonance apparatus according to claim **1**, wherein

the probe adjuster includes at least one operation unit that is able to be operated by an operator,

a characteristic of the probe is adjusted by the operation unit being operated, and

the probe adjuster is disposed in a manner that the operation unit and the opening are positioned to face each other.

3. The nuclear magnetic resonance apparatus according to claim **2**, wherein

a scale indicating a reference position for the operation unit and an amount of operation from the reference position is drawn on the probe adjuster, and

a characteristic of the probe is adjusted according to an amount of operation from a reference position for the operation unit.

4. The nuclear magnetic resonance apparatus according to claim **1**, wherein the second protective cover includes an opening and closing door configured to open and close the opening.

5. The nuclear magnetic resonance apparatus according to claim **2**, wherein the second protective cover includes an opening and closing door configured to open and close the opening.

6. The nuclear magnetic resonance apparatus according to claim **3**, wherein the second protective cover includes an opening and closing door configured to open and close the opening.

7. The nuclear magnetic resonance apparatus according to claim **1**, wherein

the sample container is inserted into the magnetic field from one side of the magnetic field generator, and

the probe is inserted into the magnetic field from another side of the magnetic field generator.

8. The nuclear magnetic resonance apparatus according to claim **7**, wherein

the second protective cover includes a removable protection member on one side of the magnetic field generator, and

the sample container is inserted into the magnetic field from one side of the magnetic field generator after the removable protection member is removed.

9. The nuclear magnetic resonance apparatus according to claim **8**, wherein

the second protective cover includes the opening on another side of the magnetic field generator.

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